

Trunk Anatomy, Embryology & Birth Defects

Comprehensive Learning Objective Guide, Practice Quiz & Rapid Review Sheet

MODULE 1: ANATOMICAL TERMINOLOGY OF THE TRUNK

Obj 1: Trunk Subdivisions

The trunk is the central body axis. Subdivided into 5 main regions: **Thorax** (chest cavity/ribcage), **Abdomen** (belly/viscera), **Pelvis** (pelvic girdle contents), **Perineum** (inferior pelvic floor/genitals), and **Back** (dorsal vertebral axis).

Quiz Pitfall: Thorax and abdomen are separated by the **diaphragm**.

MUST REMEMBER: The trunk has 5 regions: Thorax, Abdomen, Pelvis, Perineum, and Back.

Obj 2: Surface Landmarks & Regional Terms

Anterior landmarks: **Sternal** (breastbone), **Pectoral** (chest), **Mammary** (breast), **Umbilical** (navel), **Inguinal** (groin). Posterior landmarks: **Vertebral** (spine), **Scapular** (shoulder blade), **Lumbar** (lower back), **Sacral** (between hips), **Gluteal** (buttocks).

Quiz Pitfall: Inguinal = groin (anterior). Lumbar = lower back (posterior). Sacral = tailbone/hip area.

MUST REMEMBER: Memorize anterior (**Sternal, Umbilical, Inguinal**) vs posterior (**Vertebral, Lumbar, Sacral**) terms.

Obj 3: Directional Terms & Relationships

Ipsilateral = same side; **Contralateral** = opposite side. **Parietal** = lines cavity wall; **Visceral** = covers organ directly. Superior/Inferior, Anterior/Posterior, Medial/Lateral, Superficial/Deep.

Quiz Pitfall: Common mistake: Confusing parietal (wall) with visceral (organ surface).

MUST REMEMBER: Parietal = wall, Visceral = organ surface. Ipsilateral = same side, Contralateral = opposite side.

Obj 4: Planes & Cavities

Planes: **Median (Midsagittal)** = equal left/right; **Sagittal** = unequal left/right; **Coronal (Frontal)** = anterior/posterior; **Transverse (Axial)** = superior/inferior. Cavities: Dorsal (cranial/vertebral) and Ventral (thoracic & abdominopelvic, separated by diaphragm).

Quiz Pitfall: Serous membranes: Pleura (lungs), Pericardium (heart), Peritoneum (abdomen).

MUST REMEMBER: Diaphragm divides thoracic and abdominopelvic cavities. Midsagittal = equal left/right halves.

MODULE 2: FIRST WEEK OF HUMAN DEVELOPMENT

Obj 1: Ovulation to Implantation Chronology

Day 1: **Fertilization** in the ampulla of uterine tube. Days 1-3: **Cleavage** divisions (zygote to blastomeres). Day 3: **Morula** (16-cell solid ball). Day 4-5: **Blastocyst** formation (inner embryoblast + outer trophoblast). Day 6: **Implantation** begins into endometrium.

Key Concept: Fertilization occurs specifically in the **ampulla** (widest part of tube), NOT in the uterus.

MUST REMEMBER: Fertilization (Day 1, Ampulla) -> Morula (Day 3, 16 cells) -> Blastocyst (Day 4-5) -> Implantation (Day 6).

Obj 2: Sperm Conditioning & Fertilization Steps

1. **Capacitation:** 7-hour glycoprotein removal in female tract. 2. **Acrosome Reaction:** Release of enzymes to penetrate zona pellucida. 3. **Fusion:** Sperm enters oocyte, triggering cortical reaction (prevents polyspermy), completion of meiosis II, and restoration of diploidy (46 chromosomes).

Key Concept: Capacitation happens BEFORE the acrosome reaction. Without capacitation, sperm cannot penetrate the zona pellucida.

MUST REMEMBER: Capacitation (7 hrs in female tract) -> Acrosome Reaction -> Zona penetration -> Diploidy restored.

MODULE 3: SECOND TO EIGHTH WEEK (EMBRYONIC PERIOD)

Obj 1: Trophoblast Differentiating Layers

Trophoblast splits into two layers: 1. **Cytotrophoblast** (inner layer of mononucleated mitotic cells). 2. **Syncytiotrophoblast** (outer multinucleated mass without cell boundaries). Secretes **hCG** (pregnancy test marker).

Exam Focus: Syncytiotrophoblast is invasive and produces hCG. Cytotrophoblast supplies new cells by mitosis.

MUST REMEMBER: Inner = Cytotrophoblast (cellular). Outer = Syncytiotrophoblast (multinucleated, produces hCG).

Obj 2: Bilaminar Disc & Cavities

Embryoblast splits into **Bilaminar Disc**: Epiblast (dorsal columnar cells) and Hypoblast (ventral cuboidal cells). Cavities: **Amniotic Cavity** (in epiblast) and **Secondary Yolk Sac** (from hypoblast pinching off primitive yolk sac).

Exam Focus: Epiblast gives rise to ALL three germ layers during gastrulation.

MUST REMEMBER: Epiblast (dorsal) + Hypoblast (ventral) = Bilaminar Disc. Epiblast forms the entire embryo!

Obj 3: Gastrulation & Germ Layers

Gastrulation (Week 3) converts bilaminar to **trilaminar disc** via **primitive streak** invagination.

- **Ectoderm:** Nervous system, epidermis, hair, nails, adrenal medulla.
- **Mesoderm:** Muscle, bone, cartilage, dermis, blood, cardiovascular, urogenital organs.
- **Endoderm:** Epithelial lining of GI tract, respiratory tract, urinary bladder, liver, pancreas.

Exam Focus: Persistence of the primitive streak leads to **Sacroccoccygeal Teratoma** (most common newborn tumor).

MUST REMEMBER: Gastrulation = 3 germ layers formed from epiblast. Persistent streak = Sacroccoccygeal Teratoma.

Obj 4: Notochord & Neurulation

Notochord: Solid midaxial rod formed from primitive pit cells. Induces overlying ectoderm to form **Neural Plate** -> **Neural Tube** (CNS). **Neural Crest Cells** detach from folds to form PNS, autonomic ganglia, melanocytes, and adrenal medulla.

Exam Focus: Neural tube = Central Nervous System (Brain/Spinal Cord). Neural crest = Peripheral Nervous System + Melanocytes + Adrenal Medulla.

MUST REMEMBER: Notochord induces Neural Tube (CNS). Neural Crest forms PNS and Melanocytes.

Obj 5: Mesoderm Subdivisions & Somites

Mesoderm divides into: 1. **Paraxial Mesoderm** (forms **Somites** -> Sclerotome [bone/ribs], Myotome [muscle], Dermatome [dermis]). 2. **Intermediate Mesoderm** (kidneys/gonads). 3. **Lateral Plate Mesoderm** (somatic & splanchnic layers).

Exam Focus: Somite parts: Sclerotome = Vertebrae/ribs; Myotome = Skeletal muscle; Dermatome = Dermis.

MUST REMEMBER: Paraxial = Somites (Sclerotome/Myotome/Dermatome). Intermediate = Urogenital.

MODULE 4: DEVELOPMENT OF MUSCULOSKELETAL SYSTEM

Obj 1: Muscle Origins & High-Yield Exceptions

- **Skeletal Muscle:** Paraxial mesoderm (myotomes of somites).
- **Cardiac Muscle:** Splanchnic mesoderm surrounding heart tube.

- **Smooth Muscle:** Splanchnic mesoderm (GI tract/viscera) & local mesenchyme (vessels).
- **CRITICAL EXCEPTIONS:** Iris sphincter/dilator muscles, sweat glands, and mammary gland smooth muscle come from **ECTODERM!**

High Yield: High-yield exam trap: Iris muscles are ECTODERMAL, not mesodermal!

MUST REMEMBER: Most muscle = Mesoderm. Iris sphincter/dilator & sweat gland muscle = Ectoderm.

Obj 2: Epimere vs. Hypomere Differentiation

Myotomes split into two divisions by Week 5:

- **Epimere (Dorsal Division):** Deep back extensor muscles (erector spinae, multifidus). Innervated by **Dorsal Primary Rami**.
- **Hypomere (Ventrolateral Division):** Body wall & limb flexor/extensor muscles (intercostals, abdominal obliques, rectus abdominis, flexors/extensors). Innervated by **Ventral Primary Rami**.

High Yield: Epimere = Back extensors + Dorsal rami. Hypomere = Body wall/limbs + Ventral rami.

MUST REMEMBER: Epimere = Dorsal ramus (Deep back). Hypomere = Ventral ramus (Chest, Abdomen, Limbs).

MODULE 5: HUMAN BIRTH DEFECTS (TERATOLOGY)

Obj 1: Teratogens & Principles of Teratology

- **Teratogen:** Any environmental agent (chemical, drug, virus, radiation) causing congenital anomalies.
- **Critical Period: Embryonic Period (Weeks 3–8)** is most sensitive to teratogens (organogenesis phase). Weeks 1-2 = 'All-or-None' (death or complete recovery). Weeks 9-38 = functional defects, not structural malformations.

Quiz Favorite: Weeks 3-8 is the peak period for structural birth defects!

MUST REMEMBER: Peak susceptibility to teratogens = Weeks 3 to 8 (Embryonic Organogenesis Period).

Obj 2: Classic Teratogens & Presentations

- **Alcohol:** Fetal Alcohol Syndrome (FAS) - #1 cause of intellectual disability; flat philtrum, thin upper lip.
- **Thalidomide:** Limb defects - Amelia (missing limbs) & Phocomelia (seal-like limbs).
- **Rubella Virus:** Cataracts, heart defects (PDA), deafness.
- **Smoking/Nicotine:** Low birth weight, orofacial clefts (cleft lip/palate).

Quiz Favorite: Alcohol is the most common preventable cause of mental retardation.

MUST REMEMBER: Alcohol = FAS (#1 mental retardation). Thalidomide = Phocomelia. Rubella = Cataracts/Heart.

Obj 3: Chromosomal Anomalies & Single Gene Mutations

- **Numerical Anomalies** (Meiotic Nondisjunction): Down Syndrome (Trisomy 21), Turner Syndrome (45,X - webbed neck, female monosomy), Klinefelter Syndrome (47,XXY - tall male, gynecomastia).
- **Structural Anomalies** (Breaks/Deletions): Cri-du-chat (5p deletion - cat-like cry), Angelman/Prader-Willi (Ch 15 microdeletions).
- **Single Gene Mutations:** Achondroplasia (FGFR3 mutation - short-limb dwarfism).

Quiz Favorite: Turner syndrome (45,X) is the ONLY viable monosomy in humans.

MUST REMEMBER: Turner = 45,X (monosomy). Cri-du-chat = 5p deletion. Achondroplasia = Single gene FGFR3.

PART 2: HIGH-YIELD PRACTICE QUIZ

Q1. Where does fertilization normally take place?

- A) Uterine cavity
- B) Ampulla of the uterine tube
- C) Isthmus of the uterine tube
- D) Infundibulum

Correct Answer: B

Explanation: Fertilization occurs in the ampulla, the widest part of the uterine tube, within 12–24 hours after ovulation.

Q2. A newborn infant is found to have a large sacrococcygeal teratoma. This tumor is derived from remnants of which embryonic structure?

- A) Notochord
- B) Neural crest
- C) Primitive streak
- D) Hypoblast

Correct Answer: C

Explanation: Pluripotent cells of the primitive streak normally degenerate. If remnants persist, they form sacrococcygeal teratomas containing tissues from all 3 germ layers.

Q3. Which of the following structures is correctly matched with its germ layer origin?

- A) Autonomic ganglia — Mesoderm
- B) Epithelial lining of lungs — Endoderm
- C) Sphincter pupillae muscle — Mesoderm
- D) Dermis of skin — Ectoderm

Correct Answer: B

Explanation: Endoderm forms the epithelial lining of the respiratory and GI tracts. Autonomic ganglia come from neural crest (ectoderm); iris muscles come from ectoderm; dermis comes from mesoderm.

Q4. Deep extensor muscles of the back (such as the erector spinae) develop from which division of the myotome and are innervated by which nerve branch?

- A) Hypomere; Ventral primary ramus
- B) Epimere; Dorsal primary ramus
- C) Hypomere; Dorsal primary ramus
- D) Epimere; Ventral primary ramus

Correct Answer: B

Explanation: Epimere forms deep back extensor muscles innervated by dorsal primary rami. Hypomere forms body wall and limb muscles innervated by ventral primary rami.

Q5. A pregnant woman is exposed to a severe teratogen during Week 5 of gestation. What is the most likely outcome for the fetus?

- A) No effect due to the all-or-none period
- B) Major structural malformation of an organ system
- C) Minor functional defect without structural changes
- D) Spontaneous abortion with 100% certainty

Correct Answer: B

Explanation: Weeks 3–8 represent the embryonic period of maximum susceptibility to teratogens, resulting in major structural organ anomalies.

PART 3: 5-MINUTE RAPID REVIEW SHEET FOR QUIZ DAY

Concept / Term	Key Fact to Memorize for Quiz
Diaphragm	Separates Thoracic cavity from Abdominopelvic cavity.
Parietal vs. Visceral	Parietal = Cavity Wall lining. Visceral = Organ outer surface covering.
Fertilization Site	Ampulla of Uterine Tube (Day 1).
Capacitation	7-hour glycoprotein coat removal in female tract before acrosome reaction.
Morula	Solid 16-cell ball at Day 3.
Syncytiotrophoblast	Outer multinucleated trophoblast layer that secretes hCG (pregnancy test).
Epiblast	Source of ALL 3 germ layers (Ectoderm, Mesoderm, Endoderm).
Sacroccygeal Teratoma	Caused by persistence of the Primitive Streak.
Notochord	Midaxial solid rod that induces Ectoderm to form Neural Tube.
Iris Muscles Exception	Sphincter/dilator pupillae muscles originate from ECTODERM.
Epimere vs. Hypomere	Epimere = Back extensors (Dorsal ramus). Hypomere = Wall/limbs (Ventral ramus).
Teratogen Peak Window	Weeks 3 to 8 (Embryonic Period / Organogenesis).
Fetal Alcohol Syndrome	#1 preventable cause of intellectual disability (flat philtrum, thin lip).
Thalidomide	Causes Amelia (missing limbs) and Phocomelia (seal-like limbs).
Turner Syndrome	45,X — female monosomy, webbed neck, coarctation of aorta.